Predictive Model Plan

1. Model Logic (Generated with GenAI)

Pseudo code:

```
FUNCTION build_delinquency_model(dataset):

    // Step 1: Data Preparation
    // Handle missing values in numerical columns
    median_income = MEDIAN(dataset["Income"])
    FILL_MISSING_VALUES(dataset["Income"], median_income)

    median_loan_balance = MEDIAN(dataset["Loan_Balance"])
    FILL_MISSING_VALUES(dataset["Loan_Balance"], median_loan_balance)

    // Create the target variable
    dataset["Delinquency"] = 0
    FOR EACH customer_row IN dataset:
        IF "Late" OR "Missed" IN customer_row["Month_1"] to
["Month_6"]:
            SET customer_row["Delinquency"] = 1

    // Encode categorical features
    columns_to_encode = ["Employment_Status", "Credit_Card_Type",
"Location"]
    FOR EACH column IN columns_to_encode:
        one_hot_encoded_data = ONE_HOT_ENCODE(dataset[column])
        ADD one_hot_encoded_data TO dataset
        REMOVE original column FROM dataset

    // Step 2: Model Training
    // Define features (X) and target (y)
    features = ALL columns EXCEPT "Customer_ID", "Month_1"..."Month_6",
"Delinquency"
    target = "Delinquency"

    // Split data into training and testing sets
    X_train, X_test, y_train, y_test = SPLIT_DATA(features, target,
test_size=0.2, random_state=42)

    // Initialize and train the logistic regression model
    model = INITIALIZE_LOGISTIC_REGRESSION_MODEL()
    model.TRAIN(X_train, y_train)

    // Step 3: Model Evaluation
    // Make predictions on the test set
    predictions = model.PREDICT(X_test)

    // Calculate evaluation metrics
    accuracy = CALCULATE_ACCURACY(y_test, predictions)
    precision = CALCULATE_PRECISION(y_test, predictions)
    recall = CALCULATE_RECALL(y_test, predictions)
```

```
f1_score = CALCULATE_F1_SCORE(y_test, predictions)

PRINT "Model Evaluation Metrics:"
PRINT "Accuracy:", accuracy
PRINT "Precision:", precision
PRINT "Recall:", recall
PRINT "F1-Score:", f1_score

RETURN model
```

Explanation of Steps

Step 1: Data Preparation

- **Handle Missing Values:** Before we can use the data, we need to fill in the missing `Income` and `Loan_Balance` values. The pseudocode shows that we calculate the median of each of these columns and then use that value to fill in any empty cells.
- **Create the Target Variable:** This is a crucial step for any predictive model. We're creating a new column, `Delinquency`, that will be our prediction target. The pseudocode iterates through the payment history columns (`Month_1` to `Month_6`) and assigns a `1` to any customer who had at least one "Late" or "Missed" payment. If all six payments were "On-time," the value remains `0`.
- **Encode Categorical Features:** Machine learning models require numerical input. The pseudocode outlines how we transform text-based columns like `Employment_Status` into a series of new numerical columns using **one-hot encoding**.

Step 2: Model Training

- **Split the Data:** We need to split the dataset into two parts: a **training set** and a **testing set**. The training set is what the model will learn from, while the testing set is reserved to evaluate its performance on unseen data. The pseudocode suggests using 80% of the data for training and 20% for testing.
- **Initialize and Train:** The final part of this step is to instantiate a logistic regression model and then use the training data to teach the model how to make predictions.

Step 3: Model Evaluation

- **Make Predictions:** Once the model is trained, we use it to make predictions on the testing set.
- **Calculate Evaluation Metrics:** The pseudocode shows how to calculate various metrics that are essential for understanding how well the model performed. **Accuracy** tells us the overall correctness, while **Precision**, **Recall**, and the **F1-Score** provide a more nuanced view of the model's ability to correctly identify delinquent customers.

2. Justification for Model Choice

I selected the logistic regression model because it is a **simple, yet highly effective** choice for this type of problem. It's particularly well-suited for predicting a binary outcome, such as whether a customer is `Delinquent` (1) or `Not Delinquent` (0).

Here are a few key reasons for this selection:

- **Interpretability:** Logistic regression is a transparent model. Unlike more complex "black box" models, it provides clear coefficients for each variable. This allows us to understand exactly how each factor, like a customer's `Credit_Score` or `Debt_to_Income_Ratio`, influences the probability of delinquency.
- **Efficiency:** It's computationally inexpensive and fast to train, even on large datasets. This makes it a great baseline model to start with before moving on to more complex techniques.
- **Foundation:** It serves as a solid foundation. If we find that logistic regression performs well, we have an interpretable model that's easy to explain to stakeholders. If its performance is lacking, we can then use its results as a benchmark to justify the use of more complex models like Random Forests or Gradient Boosting.

3. Evaluation Strategy

. Performance Metrics and Interpretation

For this kind of classification problem, we need to go beyond simple accuracy. We'll use a combination of metrics that provide a more complete picture of the model's performance on the test data:

- **Accuracy:** This is the percentage of all predictions the model got right. While it's a good starting point, it can be misleading if the dataset is unbalanced (e.g., if only a small percentage of customers are delinquent).
- **Precision:** Of all the customers the model predicted as delinquent, how many actually were? High precision is important to avoid flagging low-risk customers, which can lead to unfair decisions.
- **Recall:** Of all the customers who were actually delinquent, how many did the model correctly identify? High recall is essential to make sure we're not missing a large number of high-risk customers.

- **F1-Score:** This is the harmonic mean of precision and recall. It's a balanced metric that's especially useful when you need to weigh both false positives and false negatives.
- **AUC (Area Under the ROC Curve):** This measures the model's ability to distinguish between delinquent and non-delinquent customers. An AUC score of 1.0 indicates a perfect model, while 0.5 suggests the model is no better than random guessing.

2. Bias Detection and Reduction

To ensure fairness, we will actively work to detect and reduce bias. We will evaluate our model's performance across different demographic subgroups, particularly age, income, and location, to make sure the model isn't unfairly targeting any specific group. If a significant performance gap is found, we might consider techniques like **disparate impact analysis** or re-weighting the training data to balance the representation of different groups.

3. Ethical Considerations

Predicting a customer's financial behavior comes with serious ethical responsibilities. We would keep the following in mind:

- **Fairness:** The model should not make decisions that disproportionately harm or benefit specific groups. It's not enough for the model to be accurate; it must also be fair.
- **Transparency:** Because we are using a transparent model like logistic regression, we can explain why the model made a certain prediction. This is crucial for accountability.
- **Human Oversight:** The model's predictions should be a tool to assist human decision-makers, not replace them. A human should always review the results before taking action.